

Ministry of Education and Science of the Russian Federation



Dmitry Gerasimov

**Adaptive and Robust Control
Practical course**

Saint-Petersburg

2020

The practical course provides recommendations for the implementation of laboratory, practical and term works on the course "Adaptive and robust control." The objective of this course is to give the exercises and brief theoretical background to the principles of adaptive and robust control of the plants in presence of incomplete information about the state and the parameters of the plant or exogenous (disturbance or reference) signal.

Each work contains basic theoretical background related to the issue of the topic, the order of calculations and simulation experiments, and the list of questions and additional tasks necessary for knowledge consolidation.

The works can be performed independently from each other as well as steps in the solution of a term or large practical works.

Content

INTRODUCTION	4
Main definitions	5
Work #1. SIMPLE EXAMPLE OF ADAPTIVE CONTROLLER DESIGN	7
Work #2. SIMPLE EXAMPLES OF ADAPTIVE ROBUST CONTROLLERS DESIGN	10
Work # 3. STATE FEEDBACK ADAPTIVE CONTROL	13
Work # 4. STATE FEEDBACK ROBUST AND ROBUST ADAPTIVE CONTROL	21
Work #5. PLANT PARAMETERIZATION (SCHEMES #1 and #2)	25
Работа #6. ADAPTIVE OBSERVER DESIGN FOR LTI PLANTS	32

INTRODUCTION

The aim of the works is the getting knowledge about the methods for the synthesis of adaptive and robust control systems based on instrumental of Lyapunov functions and the development of research skills based on the knowledge gained. In the course of the work, it is planned to study the basic algorithms of adaptive and robust control of scalar and multidimensional objects in the case of measurability and immeasurability of the state vector. Attention is paid to the plants parameterization used for linear systems identification, adaptive state observers design and model reference adaptive control as well.

It is assumed that the students passed the courses of linear and nonlinear control theory as well as the course of systems identification can begin this course.

To perform the works students should first study the theoretical background given in Lectures and the books (see References) and learn MatLab/Simulink environment.

The implementation of each work includes computer simulation, making a report (explanatory note) and defending. The defending includes the individual discussion between the teacher and the student/students, during which the quality of the knowledge and skills acquired by the student is revealed

The report consists of (mandatory):

1. Front page with names, number of work and its name;
2. Problem statement;
3. Theoretical background with necessary formulas, schemes and diagrams;
4. Experimental part with simulation and calculation results.
5. Conclusions.

Main definitions

Adaptive system is a system which continuously monitors its own performance in terms of some criteria or optimal condition and modifies its own parameters by a closed-loop action to approach this optimum behavior.

Adaptive control system is a feedback control system intelligent enough to adjust its characteristics in a changing environment so as to operate in an optimum manner according to some specified criterion

Uncertain plant is a plant (mathematical model of plant) containing unknown characteristics — parameters, functions, structures — that having the significant influence on the behavior of this plant.

Formally, the uncertainties a plant can be divided into four types:

- parametric uncertainties, implying the presence of unknown constant parameters in the plant;

- signal uncertainties representing functions of time (for example, unknown parameters in time or external disturbances);

- functional uncertainties representing unknown functions of the plant state variables;

- structural uncertainties suggesting the presence of unknown structures in the plant (for example, unmodeled dynamics of sensors or actuators).

Robust system is a system that guarantees “reasonable” closed-loop system performance in presence of parametric, signal and structural uncertainties as well as unmodeled dynamics. As a rule, a robust system does not provide a compensation of uncertainties.

Robust control is a control that insensitive or has “reasonable” sensitivity with respect to bounded plant parameters variations, “reasonable” classes of unmodelled dynamics and bounded disturbances.

List of notations

R^n is the linear space of n -dimensional real vectors;

$H(s), W(s), L(s)$ are the transfer functions (symbolic form of differential equation);

s is the Laplace operator;

$y(t) = H(s)[f(t)]$ is the transformation of the signal $f(t)$ by applying transfer function $H(s)$;

$\text{grad} \{f(x)\}$ is the gradient of the function $f(x)$;

$\|x\|, \|P\|$ is the Euclidian vector norm or the spectral matrix norm;

$\in (\notin)$ means that some element of some space belongs (does not belong) to a space.

Work #1. SIMPLE EXAMPLE OF ADAPTIVE CONTROLLER DESIGN

Problem statement. Consider the scalar plant

$$\dot{x} = \theta x + u, \quad (1.1)$$

where x is the state variable (coincides with the output), u is the control, θ is the unknown scalar parameter.

The objective is to design a control that will completely compensate for θ provide the limiting equality

$$\lim_{t \rightarrow \infty} (x_m(t) - x(t)) = \lim_{t \rightarrow \infty} \varepsilon(t) = 0, \quad (1.2)$$

where $\varepsilon = x_m - x$ is the control error, x_m is the output of the reference model

$$\dot{x}_m = -\lambda x_m + \lambda g, \quad (1.3)$$

with a piece-wise continuous reference signal g and the parameter $\lambda > 0$ responsible for the transient performance of closed-loop system after adaptation process.

Problem solution. Solution consists of the adjustable control

$$u = -\hat{\theta}x - \lambda x + \lambda g. \quad (1.4)$$

and the adaptation algorithm

$$\dot{\hat{\theta}} = -\gamma x \varepsilon \quad (1.5)$$

with adaptation gain $\gamma > 0$.

The order of the work

1. Using the parameters of the plant (1.1) and the reference (1.3) given in Table 1 design the nonadaptive controller

$$u = -\theta x - \lambda x + \lambda g.$$

Make a simulation experiment, in which the plant parameter θ is increased rapidly (controller **parameters are not changed**) so that the closed-loop system loses stability. Plot separately the variable x together with x_m and variable u .

2. Design the adaptive control (1.4), (1.5) and repeat the experiment #1 with changing plant parameters and $\hat{\theta}(0) = 0$. Adaptation gain γ is selected experimentally based on better representation of results. Plot separately the variable x together with x_m , variable u and additionally variable $\tilde{\theta} = \theta - \hat{\theta}$.

3. Make experiment for different gains γ .

4. Make a conclusion.

Table 1. Conditions for controllers design

#	Plant parameter θ	Ref. model parameter λ	Reference $g(t)$	#	Plant parameter θ	Ref. model parameter λ	Reference $g(t)$
1	6	1	$\sin 2t$	16	7	1	$\cos t + 3$
2	2	2	$\cos 4t$	17	8	2	4
3	4	3	$\text{sign}(\sin 3t) + 2$	18	9	3	$\cos 3t + 2$
4	1	4	2	19	1	4	$3\sin t$
5	5	5	$2\sin 5t$	20	2	5	$0,5\cos 8t + 1$
6	8	6	1	21	3	6	$\sin 2t + 6$
7	3	7	$7\sin t$	22	4	7	9
8	9	8	$5\sin 0,5t + 6$	23	5	8	$\cos t + 4$
9	10	9	$\text{sign}(\cos t) + 3$	24	6	9	$9\sin 0,5t + 12$
10	15	1	$\text{sign}(\sin 2t) + 5$	25	7	10	$\sin 8t + 2$
11	4	2	$\sin 5t + 0,5\cos 0,2t + 2$	26	8	1	10
12	8	3	$2\text{sign}(\sin 0,4t) + 3$	27	3	2	$\cos 2t + 2$
13	6	4	$3\sin 0,5t + 3$	28	1	3	$\text{sign}(\cos t) + 8$
14	7	5	$2\sin 0,2t + \sin 0,1t + 8$	29	2	4	$3\text{sign}(\sin t) + 4$
15	11	6	$4\text{sign}(\cos 2t) + 5$	30	3	5	$3\sin t + 10$

Problems and questions

1. Solve the problem of adaptive control for the plant

$$\dot{x} = \theta_1 x^5 + \theta_2 u,$$

where θ_1, θ_2 are unknown parameters ($\theta_2 > 0$). Objective is defined by the equality (1.2).

2. Solve the problem of adaptive control for the plant

$$\dot{x} = -x + \theta_1 u + \theta_2,$$

where θ_1, θ_2 are unknown parameters ($\theta_1 > 0$). Objective is defined by the equality (1.2).

3. Solve the problem of adaptive control for the plant

$$\dot{x} = \theta x^3 + u,$$

where θ is unknown parameter. Objective is defined by the equality (1.2).

4. Solve the problem of adaptive control for the plant

$$\dot{x} = \theta(t)x + u,$$

where $\theta(\omega) = c_0 + c_1\omega + c_2\omega^2$, c_0, c_1, c_2 are unknown coefficients, ω is a known function of time. Objective is defined by the equality (1.2).

5. Solve the problem of adaptive control for the plant

$$\dot{x} = \theta_1 x + u + \theta_2 \sin \omega t,$$

where θ_1, θ_2 are unknown parameters, ω is the known constant frequency. Objective is defined by the equality (1.2).

6. Solve the problem of adaptive control for the plant

$$\dot{x} = \theta_1 x + u + \theta_2 x^2,$$

where θ_1, θ_2 are unknown parameters. Objective is defined by the equality (1.2).

7. Solve the problem of adaptive control for the plant

$$y = \frac{K}{Ts+1}[u],$$

where K, T are unknown parameters ($K > 0$). Objective is defined by the equality (1.2).

8. Find an adaptation algorithm for the problem above by applying the Lyapunov function

$$V(\varepsilon, \tilde{\theta}) = |\varepsilon| + \frac{1}{2} \tilde{\theta}^2.$$

9. Formulate the conditions for asymptotic and exponential stability using Lyapunov functions.

10. Does the solution of the adaptive control problem above ensures the parametric error convergence to zero?

Work #2. SIMPLE EXAMPLES OF ADAPTIVE ROBUST CONTROLLERS DESIGN

Problem statement. Consider disturbed plant

$$\dot{x} = \theta x + u + \delta, \quad (2.1)$$

where δ is unknown bounded disturbance such that $|\delta(t)| \leq \bar{\delta}$. As in the work #1, x is the state variable, u is the control, θ is the unknown constant parameter.

The objective is to design a control that will provide the boundness of all the signals in the closed-loop system and ensure the inequality

$$|x_m(t) - x(t)| \leq \Delta, \quad \forall t \geq T, \quad (2.2)$$

where Δ , T are the maximum steady-state error and transient time, respectively, $x_m(t)$ is the output of reference model (1.2). It is assumed that the constants Δ and T can be regulated by the parameters of controller.

Solution # 1. Solution obtained by applying nonlinear static feedback

$$\hat{\theta} = -\gamma x \varepsilon \quad (2.3)$$

in (1.4) instead of adaptation algorithm (1.5). Substituting (2.3) for $\hat{\theta}$ in (1.4) we obtain **robust** control law

$$u = \gamma x^2 \varepsilon - \lambda x + \lambda g. \quad (2.5)$$

Solution # 2. Modify adaptation algorithm (1.5) by applying the leakage factor (static linear feedback):

$$\dot{\hat{\theta}} = -\sigma \hat{\theta} - \gamma x \varepsilon, \quad (2.8)$$

where σ is the constant value. Then use this **adaptive robust** modification together with adjustable control (1.4).

The order of the work

1. Based on the parameters given in Table.1 make experiment with the system closed by control (1.4) and the adaptation algorithm (1.5) for the plant (2.1) disturbed by the signal [3]

$$\delta(t) = (1+t)^{-1/8} \left[1 - \theta(1+t)^{-1/4} - \frac{3}{8}(1+t)^{-5/4} \right].$$

The controller parameters are given as $\gamma = 0.25$, $x(0) = 1$ и $\hat{\theta}(0) = 1$. Reference $g(t)$ equals zero. Separately plot variable x together with x_m , variable u and variable $\tilde{\theta} = \theta - \hat{\theta}$. Make a conclusion.

2. Repeat the previous experiment with the control (2.5) and three different gains γ . Reference signals $g(t)$ are taken from Table.1. Define the influence of the gain γ on the closed-loop system properties. Separately plot variable x together with x_m and variable u .

3. Repeat experiments 2.2 using the control (1.4) with σ -modification (2.8) and with the same three different γ . Repeat experiment for a certain γ and three different coefficients σ . Reference signals $g(t)$ are taken from Table.1. Define the influence of the coefficients γ and σ on the closed-loop system properties. Separately plot variable x together with x_m , variable u and variable $\tilde{\theta} = \theta - \hat{\theta}$.

Problems and questions

1. Show that the robust controllers presented in the work ensure the stability of the closed-loop system despite the changes of the parameter θ .

2. Is an asymptotically stable system robust with respect to external disturbances? Please, clarify.

3. Is an exponentially stable system robust with respect to external disturbances? Please, clarify.

4. Is the maximum steady-state control error increased by increasing σ ? Please, clarify.

5. Is the maximum steady-state control error decreased by increasing γ ? Please, clarify.

6. Solve the problem of robust adaptive control (Solution of the second type) for the plant

$$\dot{x} = x^4 + \theta x + u,$$

where θ is unknown parameter. Objective is defined by the inequality (2.2).

7. Prove the stability of the robust system with the plant (2.1), control (1.4) and nonlinear static feedback

$$\hat{\theta} = -\gamma x \text{sign}(\varepsilon)$$

with a positive adaptation gain γ .

8. Prove the stability of the robust system with the plant (2.1), control (1.4) and the σ -modification of adaptation law

$$\dot{\hat{\theta}} = -\sigma \hat{\theta} - \gamma x \text{sign}(\varepsilon),$$

where γ, σ are positive constants.

9. Solve the problem of robust adaptive control of the plant

$$\dot{x} = \theta(x^2 + 1) + u + \delta,$$

where θ is unknown constant parameter, δ is the bounded disturbance such that $|\delta(t)| \leq \bar{\delta}$. Objective is defined by the inequality (2.2).

10. Solve the problem of robust adaptive control of the plant

$$\dot{x} = \theta \sin(x) + u + \delta,$$

where θ is unknown constant parameter, δ is the bounded disturbance such that $|\delta(t)| \leq \bar{\delta}$. Objective is defined by the inequality (2.2).

Work # 3. STATE FEEDBACK ADAPTIVE CONTROL

Problem statement. Consider the plant presented in the “state-space” form

$$\dot{x} = Ax + bu, \quad x(0) \quad (3.1)$$

$$y = c^T x, \quad (3.2)$$

where $x \in R^n$ is the state vector, $u \in R$ is the control variable, $y \in R$ is the regulated variable,

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ b_0 \end{bmatrix}, \quad c = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

$a_i, i = \overline{0, n-1}$ are unknown parameters, b_0 is the coefficient which is known for the sake of simplicity.

The problem is to compensate for the parametric uncertainty and provide the limiting equality

$$\lim_{t \rightarrow \infty} \|x_M(t) - x(t)\| = \lim_{t \rightarrow \infty} \|e(t)\| = 0, \quad (3.3)$$

where $e = x_M - x$ is the vector of control errors, $x_M \in R^n$ is the state vector of reference model

$$\dot{x}_M = A_M x_M + b_M g, \quad (3.4)$$

$$y_M = c_M^T x_M \quad (3.5)$$

with piece-wise continuous reference signal $g(t)$ and matrices

$$A_M = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_{M0} & -a_{M1} & -a_{M2} & \cdots & -a_{Mn-1} \end{bmatrix}, \quad b_M = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ a_{M0} \end{bmatrix}, \quad c_M = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

The parameters $a_{Mi}, i = \overline{1, n-1}$ are calculated based on the pole-placement approach [1, 2] to provide the desired transient performance for the closed-loop system after adaptation process.

Note that in general case, the class of plants is restricted by the following assumption.

Assumption (Matching conditions). There is an n - dimensional vector θ and a scalar κ such that the matrices A , b , A_M and b_M are related via equalities

$$A_M = A + b\theta^T, \quad b = \kappa b_M \quad (3.6)$$

called matching conditions.

This assumption permits to apply *certainty equivalence principle* and design an error model used for derivation of an adaptation algorithm.

Problem solution. Assuming the plant parameters known we design a control providing condition (3.3) with desired transient performance determined by transient time t_n and the overshooting $\bar{\sigma}$.

To this end we define the control error as $e = x_M - x$ and calculate its time derivative in view of (3.1), (3.4) and (3.6):

$$\begin{aligned} \dot{e} &= \dot{x}_M - \dot{x} = A_M x_M + b_M g - Ax - bu = \\ &= \underline{A_M} x_M + \frac{1}{\kappa} \underline{b} g - (\underline{A_M} - \underline{b}\theta^T)x - \underline{b}u = A_M e + b \left(\theta^T x - u + \frac{1}{\kappa} g \right), \end{aligned} \quad (3.7)$$

where $\theta^T = [\theta_1, \theta_2, \dots, \theta_n]$ is the vector of constant parameters defined by the difference between matrices A and A_M :

$$\theta_1 = \frac{-a_{M0} + a_0}{b_0}, \quad \theta_2 = \frac{-a_{M1} + a_1}{b_0}, \quad \dots, \quad \theta_n = \frac{-a_{Mn-1} + a_{n-1}}{b_0}, \quad \kappa = \frac{b_0}{a_{M0}}.$$

Expression (3.7) is reduced to the equality

$$\dot{e} = A_M e + b \left(\theta^T x - u + \frac{1}{\kappa} g \right), \quad (3.8)$$

which can be used for design of nonadaptive control

$$u = \theta^T x + \frac{1}{\kappa} g. \quad (3.9)$$

Replacing (3.9) in (3.8) we obtain the control error model ensuring exponential convergence of the error e to zero:

$$\dot{e} = A_M e.$$

Since initially the parameters of the matrix A are unknown, the control law (3.9) is not implementable. Therefore replace the unknown parameters θ by their estimates $\hat{\theta}$ in (3.9) and obtain the adjustable control law

$$u = \hat{\theta}^T x + \frac{1}{\kappa} g. \quad (3.10)$$

Substitute the last equality in (3.8) and get the control error model

$$\dot{e} = A_M e + b \tilde{\theta}^T x, \quad (3.11)$$

where $\tilde{\theta} = \theta - \hat{\theta}$ is the vector of parametric errors.

Generalizing solution presented in the work #1 for multivariable case we select the Lyapunov function candidate

$$V = \frac{1}{2} e^T P e + \frac{1}{2\gamma} \tilde{\theta}^T \tilde{\theta},$$

where $P = P^T \succ 0$ is the solution of the Lyapunov equation

$$A_M^T P + P A_M = -Q \quad (3.12)$$

with preselected positively defined constant matrix Q . Calculating time derivative of V in view of (3.11) we obtain:

$$\dot{V} = -\frac{1}{2} e^T Q e + \tilde{\theta}^T x b^T P e + \frac{1}{\gamma} \tilde{\theta}^T \dot{\tilde{\theta}}.$$

Now, if we select the following structure for the adaptation algorithm

$$\dot{\hat{\theta}} = \gamma x b^T P e, \quad \hat{\theta}(0) = 0 \quad (3.13)$$

then the Lyapunov function derivative will be nonpositive, i.e.

$$\dot{V} = -\frac{1}{2} e^T Q e \leq 0.$$

As a result, the choice (3.13) ensures the objective (3.3).

Note that the coefficient $\gamma > 0$ in (3.13) is called *adaptation gain* responsible for the rate of parametric convergence (3.10).

Thus the algorithm of adaptive control consists of adjustable control law (3.10) and adaptation algorithm (3.13), in which the matrix P is calculated using (3.12).

For any initial conditions in the plant and the adaptation algorithm $x(0)$, $\hat{\theta}(0)$ and any bounded g adaptive control (3.10), (3.13) ensures [6, 7]:

C.1. the boundedness of all the signals in the closed-loop system;

C.2. asymptotic convergence of the control error e to zero;

C.3. exponential convergence of the vector $\hat{\theta}$ to the vector θ if x satisfies the *persistent excitation condition*

$$\int_t^{t+T} x(\tau) x^T(\tau) d\tau > \alpha I \quad (3.14)$$

for some constant values $\alpha > 0$, $T > 0$.

Condition (3.14) is equivalent to existence at least $(n+1)/2$ harmonics (spectral lines) in the vector x . Note that in the framework of this problem the frequency properties of x are dependent from the external signal g . As a result the persistent excitation condition can be reformulated in terms of signal g ;

C.4. if the vector x satisfies the condition (3.14), then there is an optimal coefficient γ , for which the rate of parametric convergence of $\tilde{\theta}$ to zero is maximum.

The order of the work

1. Using the desired transient time t_n and maximum overshooting $\bar{\sigma}$ given in Table 3 and the pole-placement approach [1, 2] design reference model (3.4), (3.5). If the maximum overshooting is zero, it worth selecting binomial polynomial of the second order (since the plant of the second order and relative degree)

$$s^2 + 2\omega_0 s + \omega_0^2, \quad (3.16)$$

where ω_0 is the radius of the poles distribution calculated as

$$\omega_0 = \frac{t_n'}{t_n}$$

with normalized transient time of the system (here we assume $\omega_0' = 1$)

$$h(t) = \frac{1}{s^2 + 2s + 1} [1(t)].$$

If the maximum overshooting is zero, Butterworth polynomial

$$s^2 + \sqrt{2}\omega_0 s + \omega_0^2$$

is to be selected. In this case the normalized transient time for calculation of ω_0 is defined by the system

$$h(t) = \frac{1}{s^2 + \sqrt{2}s + 1} [1(t)] \quad (3.17)$$

To calculate the coefficients of the matrix A_M and the vector b_M ($n=2$) of the reference model we compare the companion polynomial of A_M with (3.16) or (3.17) depending on the choice.

Plot step response function of the reference model and show t_n and the overshooting $\bar{\sigma}$;

2. Based on assumption that the plant parameters are known design and simulate the closed-loop system with nonadaptive control (3.9) with

$$\theta_1 = \frac{-a_{M0} + a_0}{b_0}, \theta_2 = \frac{-a_{M1} + a_1}{b_0}.$$

Make three experiments with three sets of **plant** parameters:

- a_0 and a_1 ;
- negligibly change a_0 and a_1 parameters so that the system preserves the stability;
- change the parameters so that the system loses the stability.

Separately plot the elements of vector $x(t)$ together with the corresponding elements of $x_M(t)$. Plot the elements of $e(t)$.

3. Repeat this triple of experiments with adaptive controller (3.10) and the adaptation algorithm (3.13) instead of (3.9). Demonstrate properties 1-4 via simulation.

Make experiment with original parameters of the plant and three values of γ ;

Repeat the experiment for $g(t) = 1$.

Separately plot the elements of vector $x(t)$ together with the corresponding elements of $x_M(t)$. Plot the elements of $e(t)$. Plot the elements of $\tilde{\theta} = \theta - \hat{\theta}$.

4. Make conclusions for all the experiments.

Table 3 The parameters of the plant and the reference model quantities

#	Matrix A	Coeff. b_0	Transient time, t_n	Maximum overshooting $\bar{\sigma}$, %	Reference $g(t)$
1	$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$	1	0,16	0	$2\text{sign}(\cos t) + 3$
2	$\begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$	2	0,3	0	$\cos t + 3\sin 2t + 5$
3	$\begin{bmatrix} 0 & 1 \\ 4 & 0 \end{bmatrix}$	3	0,9	0	$\text{sign}(\cos 2t)$
4	$\begin{bmatrix} 0 & 1 \\ 1 & 4 \end{bmatrix}$	4	0,2	0	$\text{sign}(\sin 0,5t) + 2$
5	$\begin{bmatrix} 0 & 1 \\ -2 & 5 \end{bmatrix}$	5	0,6	0	$10\cos 0,5t + 2\sin t + 12$
6	$\begin{bmatrix} 0 & 1 \\ 9 & -1 \end{bmatrix}$	6	0,3	0	$\text{sign}(\sin 0,5t) + 3$

7	$\begin{bmatrix} 0 & 1 \\ -2 & 3 \end{bmatrix}$	7	0,7	0	$0,5\text{sign}(\sin 0,7t)+1$
8	$\begin{bmatrix} 0 & 1 \\ 2 & 2 \end{bmatrix}$	8	0,1	0	$5\sin 0,5t+4\cos 0,1t+8$
9	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	9	0,9	0	$\text{sign}(\cos t)+3$
10	$\begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}$	1	3,5	15	$\text{sign}(\sin 2t)$
11	$\begin{bmatrix} 0 & 1 \\ -1 & 6 \end{bmatrix}$	2	0,6	15	$\sin 5t+0,5\cos 0,2t+2$
12	$\begin{bmatrix} 0 & 1 \\ 10 & 1 \end{bmatrix}$	3	0,9	15	$2\text{sign}(\sin 0,4t)+3$
13	$\begin{bmatrix} 0 & 1 \\ -1 & 5 \end{bmatrix}$	4	0,4	15	$3\text{sign}(\sin 0,5t)+3$
14	$\begin{bmatrix} 0 & 1 \\ 9 & 0 \end{bmatrix}$	5	0,2	15	$2\sin 0,2t+\sin 0,1t+8$
15	$\begin{bmatrix} 0 & 1 \\ -9 & 1 \end{bmatrix}$	6	0,5	15	$4\text{sign}(\cos 2t)+5$
16	$\begin{bmatrix} 0 & 1 \\ -7 & 6 \end{bmatrix}$	7	0,5	15	$3\text{sign}(\cos 0,2t)+3$
17	$\begin{bmatrix} 0 & 1 \\ -1 & 7 \end{bmatrix}$	8	0,45	15	$\text{sign}(\sin 0,5t)+2$
18	$\begin{bmatrix} 0 & 1 \\ -10 & 6 \end{bmatrix}$	9	0,15	15	$0,8\sin 2t+\cos 0,8t+2$
19	$\begin{bmatrix} 0 & 1 \\ 2 & 2 \end{bmatrix}$	1	0,7	15	$\text{sign}(\sin 0,3t)+1,5$
20	$\begin{bmatrix} 0 & 1 \\ -15 & 6 \end{bmatrix}$	2	1,2	0	$7\text{sign}(\cos 0,9t)+8$
21	$\begin{bmatrix} 0 & 1 \\ 11 & 1 \end{bmatrix}$	3	1,5	0	$0,4\sin 3t+\cos 0,1t+1,5$
22	$\begin{bmatrix} 0 & 1 \\ 3 & 3 \end{bmatrix}$	4	0,8	0	$6\text{sign}(\sin 0,1t)+9$
23	$\begin{bmatrix} 0 & 1 \\ 12 & 6 \end{bmatrix}$	5	0,9	0	$2\text{sign}(\sin t)+4$
24	$\begin{bmatrix} 0 & 1 \\ 0 & 11 \end{bmatrix}$	6	0,2	0	$9\sin 0,2t+9\cos 0,1t+15$
25	$\begin{bmatrix} 0 & 1 \\ 13 & 2 \end{bmatrix}$	7	0,5	0	$4\text{sign}(\sin 6t)+5$

26	$\begin{bmatrix} 0 & 1 \\ 15 & 6 \end{bmatrix}$	8	0,9	0	$4\text{sign}(\cos t) + 3$
27	$\begin{bmatrix} 0 & 1 \\ 13 & -1 \end{bmatrix}$	9	1,3	0	$\sin 0,1t + \cos 5t + 2$
28	$\begin{bmatrix} 0 & 1 \\ 4 & 4 \end{bmatrix}$	1	1,6	0	$9\text{sign}(\sin 0,1t) + 12$
29	$\begin{bmatrix} 0 & 1 \\ 9 & 7 \end{bmatrix}$	2	0,75	15	$3\text{sign}(\sin 4t) + 8$
30	$\begin{bmatrix} 0 & 1 \\ -1 & 9 \end{bmatrix}$	3	0,65	15	$7\sin 0,3t + 8\cos t + 20$

Problems and questions

1. Does the condition (3.14) hold for the case of scalar plant in work #1 if $g(t)=1$?
2. Can the solution of Lyapunov equation (3.12) be negatively definite matrix P ? Please clarify.
3. Can the solution of Lyapunov equation (3.12) be a diagonal matrix P ? Please clarify.
4. For which γ the adaptive system is unstable?
5. Give an example of 2-dimensional vector $x(t)$ not satisfying the persistent excitation condition (3.14).
6. Solve the problem of stabilization ($g(t) \equiv 0$) for the plant

$$\dot{x}_1 = x_2,$$

$$\dot{x}_2 = \theta_1 x_1 + \theta_2 x_1^3 + u,$$

where $\theta_1 = 2$, $\theta_2 = 5$ are unknown parameters. For the reference model use zero overshooting and transient time $t_n = 1\text{sec}$.

7. Solve the control problem for the plant

$$\dot{x}_1 = x_2,$$

$$\dot{x}_2 = \theta_1 x_1 x_2 + \theta_2 (x_1^2 + 1) + u,$$

where $\theta_1 = 2$, $\theta_2 = 5$ are unknown parameters. The objective is defined by equality (3.3). For the reference model use zero overshooting and transient time $t_n = 1\text{sec}$. The reference signal is given as $g = 2\text{sign}(\sin t) + 1$.

8. Solve the control problem for the plant

$$\begin{aligned}\dot{x}_1 &= x_2, \\ \dot{x}_2 &= \theta x_1 + u,\end{aligned}$$

where θ is unknown parameter . The objective is defined by equality (3.3). For the reference model use zero overshooting and transient time $t_n = 1\text{sec}$. The reference signal is given as $g = \sin t + 2$.

9. Solve the control problem for the plant

$$\begin{aligned}\dot{x}_1 &= 3x_2, \\ \dot{x}_2 &= \theta_1 x_1 + \theta_2 x_2 + u,\end{aligned}$$

where $\theta_1 = 1$, $\theta_2 = 2$ are unknown parameters. The objective is defined by equality (3.3). For the reference model use zero overshooting and transient time $t_n = 1\text{sec}$. The reference signal is given as $g = 3\text{sign}(\sin 2t) + 2$.

10. Solve the control problem for the plant

$$\begin{aligned}\dot{x}_1 &= x_2, \\ \dot{x}_2 &= \theta_1 x_1 + \theta_2 x_2 + \theta_3 u,\end{aligned}$$

where $\theta_1, \theta_2, \theta_3 \geq \theta_{03} > 0$ are unknown coefficients. The objective is defined by equality (3.3). For the reference model use zero overshooting and transient time $t_n = 1\text{sec}$. The reference signal is given as $g = \text{sign}(\sin t) + 2$.

Work # 4. STATE FEEDBACK ROBUST AND ROBUST ADAPTIVE CONTROL

Problem statement. Consider the disturbed plant

$$\dot{x} = Ax + bu + \delta, \quad x(0) \quad (4.1)$$

$$y = c^T x, \quad (4.2)$$

where δ is the vector of disturbance signals such that $\|\delta(t)\| \leq \bar{\delta}$, $x \in R^n$ is the state vector, y is the regulated variable, A , b and c are identical to the corresponding matrices of plant (3.1), (3.2).

The objective is to design a control law that provides the boundedness of all the signals in the closed-loop system and the convergence of the tracking error to residual set in accordance to the inequality

$$\|x_M(t) - x(t)\| \leq \Delta, \quad \forall t \geq T, \quad (4.3)$$

where Δ , T are the maximum steady-state error and the transient time, respectively, $x_M(t) \in R^n$ is the state vector of reference model (3.4), (3.5).

Consider two solutions of the problem based on scalar results of work #2 and the adaptive control developed in work #3.

Solution #1. Solution uses adjustable control (3.10) and modification of (3.13) called nonlinear static feedback and presented as

$$\hat{\theta} = \gamma x b^T P e, \quad (4.4)$$

where $\gamma > 0$ is the gain of feedback, P is the matrix solution of Lyapunov equation (3.12), $e = x_M(t) - x(t)$ is the control error.

The analysis of the closed-loop system can be done using the control error (3.11) and the Lyapunov function candidate

$$V = \frac{1}{2} e^T P e. \quad (4.5)$$

Based on analysis it can be shown that the system closed by robust control (3.10), (4.4) offers the following properties. For any initial conditions $e(0)$ the control law provides:

- P1.1. Boundedness of all the signals in the closed-loop system;
- P1.2. Exponential convergence of the norm $\|e\|$ to residual set including zero origin $e^* = 0$. The radius of the set can be decreased by increasing gain γ ;
- P1.3. Even in the disturbance-free case the steady state control error is not zero.

Note that replacing (4.4) into (3.10) we explicitly obtain the high-gain term $x^T x$ that does not provide the controller tuning, but suppress the influence of both uncertainty θ and the disturbance δ .

Solution #2. This solution also uses (3.10) together with σ – modification of (3.13) given by

$$\dot{\hat{\theta}} = -\sigma\hat{\theta} + \gamma x b^T P e, \quad (4.6)$$

where $\gamma > 0$ is the adaptation gain, $\sigma > 0$ is the static feedback gain, P is the matrix solution of (3.12).

The analysis of the closed-loop system can be done using the control error (3.11) and the Lyapunov function candidate

$$V = \frac{1}{2} e^T P e + \frac{1}{2\gamma} \tilde{\theta}^T \tilde{\theta},$$

where $\tilde{\theta} = \theta - \hat{\theta}$ is the vector of parametric errors.

It can be shown via analysis that the adaptive robust controller (3.10), (4.6) for any initial conditions $e(0)$ provides:

P2.1. Boundedness of all the signals in the closed-loop system;

P2.2. Exponential convergence of the norms $\|e\|$ and $\|\tilde{\theta}\|$ to residual sets including origins $e^* = 0$ and $\tilde{\theta}^* = 0$, respectively. Radius of the sets can be reduced by increasing γ and decreasing σ ;

P2.3. In general case steady state values of e and $\tilde{\theta}$ are not zero, however they can be decreased by decreasing σ .

The order of the work

1. Based on the results obtained in work #3 modify adaptation algorithm (3.13) and design robust control (3.10), (4.4) and adaptive robust control (3.10), (4.6). Design and simulate the obtained closed-loop systems using the following disturbance vector:

$$\delta(t) = [0, 6\sin 10t + 0, 1\sin 50t, \quad 0, 5\cos 12t + 0, 2\sin 30t]^T.$$

2. Make experiments for robust control (3.10), (4.4) for three different gains γ assuming $\|\delta(t)\| \equiv 0$.

Separately plot $x(t)$ together with $x_M(t)$ and $e(t)$.

3. Make experiments for robust adaptive control (3.10), (4.6) for a certain parameter σ and two different γ used in item#2 with and without $\delta(t)$ given by (4.7). Decrease σ and repeat this experiment for one of the selected gain γ .

Separately plot $x(t)$ together with $x_M(t)$, $e(t)$ and $\tilde{\theta}(t)$.

4. Make conclusions for all comparable result.

Problems and questions

1. Prove Property P1.2.

2. Prove Property P2.2.

3. For what purpose is the parameter σ used in the algorithm (4.6)?

4. Please, clarify why steady state control errors will be different from zero in the systems closed by control laws (3.10), (4.4) and (3.10), (4.6) in disturbance-free case?

5. If we increase simultaneously the parameters γ and σ in the system with (3.10), (4.6), how will it change the radius of residual set for the control error norm $\|x_M(t) - x(t)\|$?

6. Will the closed-loop system with robust control (3.10), (4.4) be stable if γ is positive, matrix A_M is Hurwitz and is given in canonical form

$$A_M = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_{M0} & -a_{M1} & -a_{M2} & \cdots & -a_{Mn-1} \end{bmatrix},$$

but the matrix P is:

$$\text{a) } P = \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}; \text{ б) } P = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}; \text{ в) } P = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}; \text{ г) } P = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}?$$

7. Solve the problem of robust control for pendulum

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= \theta_1 \sin(x_1) + \theta_2 x_2 + u + \delta, \end{aligned}$$

where θ_1 , θ_2 are the unknown parameters, δ is the bounded but unknown disturbance satisfying condition $|\delta(t)| \leq \bar{\delta}$. Objective is to design a control providing inequality (4.3) with all closed-loop signals bounded.

8. Solve the problem of robust adaptive control for the plant

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= \theta + x_2 + u + \delta, \end{aligned}$$

where θ is unknown parameter, δ is the bounded but unknown disturbance satisfying condition $|\delta(t)| \leq \bar{\delta}$. Objective is to design a control providing inequality (4.3) with all closed-loop signals bounded.

9. Repeat one of the experiments requested in the work, where instead of adaptation algorithm (4.6) apply its modification

$$\dot{\hat{\theta}} = \begin{cases} -\sigma \hat{\theta} + \gamma x k^T P e, & \text{если } \|\hat{\theta}\|_2 \geq \theta_0 \\ \gamma x k^T P e, & \text{если } \|\hat{\theta}\|_2 < \theta_0, \end{cases}$$

where $\theta_0 = 1.5\|\theta\|_2$ is some *a priori* information about uncertainty. Coefficients γ and σ should be taken from item#3 of the work order, while the disturbance should be taken from item#1.

Compare with results obtained in the work. Clarify the advantages of this scheme.

10. Solve the problem of robust adaptive control for the plant

$$\begin{aligned} \dot{x}_1 &= 2x_2, \\ \dot{x}_2 &= \theta_1 x_1 + \theta_2 x_2 + u + \delta, \end{aligned}$$

where θ_1, θ_2 are the unknown parameters, δ is the bounded but unknown disturbance satisfying condition $|\delta(t)| \leq \bar{\delta}$. Objective is to design a control providing inequality (4.3) with all closed-loop signals bounded.

Work #5. PLANT PARAMETERIZATION (SCHEMES #1 and #2)

Under plant parameterization we mean a representation of this plant in a form where its parameters are lumped together and separated from signals. Basically, such representations are useful for parameter estimation, observers design and control problems solutions. For the class of LTI plants and for several classes of nonlinear systems so called linear parameterization or linear regression is widely used and corresponds to the output or state representation of the plant as a superposition of plant parameters multiplied by some measurable functions (regressors).

Assumption. For the sake of brevity we assume all the initial conditions in the plants considered equal to zero.

To clarify the principle of plant parameterization we consider a simple problem. Consider the scalar plant

$$\begin{cases} \dot{x} = -ax + bu, & x(0) \\ y = x, \end{cases} \quad (5.1)$$

where a, b are the plant parameters, x is the state variable, u, y are the input and the output, respectively. The problem is to represent the output y in the form of linear regression

$$y = \theta^T \omega, \quad (5.2)$$

where θ is the vector of constant parameters depending on a and b , ω is the vector of functions.

Apply to (5.1) the operator of transfer function

$$H(s) = \frac{1}{s+k},$$

where s is the Laplace operator, k is a positive constant, and get:

$$\frac{1}{s+k}[\dot{x}] = -\frac{1}{s+k}[ax] + \frac{1}{s+k}[bu].$$

Making simple algebraic transformation we have:

$$\begin{aligned} \frac{s}{s+k}[x] &= -a\frac{1}{s+k}[x] + b\frac{1}{s+k}[u], \\ \frac{s \pm k}{s+k}[x] &= -a\frac{1}{s+k}[x] + b\frac{1}{s+k}[u], \\ x &= \frac{k}{s+k}[x] - a\frac{1}{s+k}[x] + b\frac{1}{s+k}[u] = (k-a)\frac{1}{s+k}[x] + b\frac{1}{s+k}[u]. \end{aligned}$$

Now introduce notations:

$$\xi_0 = \frac{1}{s+k} [x], \quad v_0 = \frac{1}{s+k} [u].$$

Replacing x by y we finally get regression (5.2):

$$y = (k-a)\xi_0 + bv_0 = \theta^T \omega,$$

where $\theta = [k-a, b]^T$, $\omega = [\xi_0, v_0]^T$.

Arguing in a similar way, we can design a parameterized representation for the n th order plant represented in the form of differential equation (“input-output” form)

:

$$y^{(n)} + a_{n-1}y^{(n-1)} + a_{n-2}y^{(n-2)} + \dots + a_0y = b_mu^{(m)} + b_{m-1}u^{(m-1)} + \dots + b_0u \quad (5.3)$$

with the parameters $a_i, i = \overline{0, n-1}, b_j, j = \overline{0, m}$.

Now introduce the stable transfer function

$$H(s) = \frac{1}{K(s)} = \frac{1}{s^n + k_{n-1}s^{n-1} + k_{n-2}s^{n-2} + \dots + k_0}$$

with the constant parameters $k_i, i = \overline{0, n-1}$.

Assuming initial conditions $y^{(i)}(0), i = \overline{0, n-1}$ zero we apply the operator $H(s)$ to (5.3). After algebraic transformations we have:

$$y = \sum_{i=0}^{n-1} (k_i - a_i) \frac{s^i}{K(s)} [y] + \sum_{i=0}^m b_i \frac{s^i}{K(s)} [u]$$

or

$$y = \theta^T \omega, \quad (5.4)$$

where

$$\theta^T = [k_0 - a_0, k_1 - a_1, \dots, k_{n-1} - a_{n-1}, b_0, b_1, \dots, b_m],$$

$$\omega^T = \left[\frac{1}{K(s)} [y], \frac{s}{K(s)} [y], \dots, \frac{s^{n-1}}{K(s)} [y], \frac{1}{K(s)} [u], \frac{s}{K(s)} [u], \dots, \frac{s^m}{K(s)} [u] \right].$$

Now parameterize the state of the plant presented in a “state-space” form

$$\begin{cases} \dot{x} = Ax + bu, & x(0), \\ y = c^T x, \end{cases} \quad (5.5)$$

where matrices are presented in canonical forms

$$A = \begin{bmatrix} -a_{n-1} & 1 & \cdots & 0 \\ -a_{n-2} & 0 & & 0 \\ \vdots & & \ddots & 1 \\ -a_0 & 0 & & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ b_m \\ \vdots \\ b_0 \end{bmatrix}, \quad c^T = [1 \ 0 \ \cdots \ 0].$$

Assuming initial vector $x(0)$ equal to zero we apply to (5.5) the transfer matrix operator

$$\Phi(s) = (Is - A_0)^{-1},$$

where I is the $n \times n$ identity matrix, A_0 is a Hurwitz matrix presented in canonical form

$$A_0 = \begin{bmatrix} -k_{n-1} & 1 & \cdots & 0 \\ -k_{n-2} & 0 & & 0 \\ \vdots & & \ddots & 1 \\ -k_0 & 0 & & 0 \end{bmatrix}.$$

Making algebraic matrix transformations we obtain:

$$\begin{aligned} (Is - A_0)^{-1}[\dot{x}] &= (Is - A_0)^{-1}A[x] + (Is - A_0)^{-1}b[u], \\ (Is - A_0)^{-1}s[x] &= (Is - A_0)^{-1}A[x] + (Is - A_0)^{-1}b[u], \\ (Is - A_0)^{-1}(s \pm A_0)[x] &= (Is - A_0)^{-1}A[x] + (Is - A_0)^{-1}b[u], \\ (Is - A_0)^{-1}(s - A_0)[x] + (Is - A_0)^{-1}A_0[x] &= (Is - A_0)^{-1}A[x] + (Is - A_0)^{-1}b[u], \\ x &= -(Is - A_0)^{-1}A_0[x] + (Is - A_0)^{-1}A[x] + (Is - A_0)^{-1}b[u], \\ x &= (Is - A_0)^{-1}(A - A_0)[x] + (Is - A_0)^{-1}b[u]. \end{aligned} \tag{5.6}$$

Taking into account the structures of matrices A , A_0 and the equality $y = x_1$ (in view of c^T) we have:

$$(A - A_0)[x] = \left(- \begin{bmatrix} -k_{n-1} & 1 & \cdots & 0 \\ -k_{n-2} & 0 & & 0 \\ \vdots & & \ddots & 1 \\ -k_0 & 0 & & 0 \end{bmatrix} + \begin{bmatrix} -a_{n-1} & 1 & \cdots & 0 \\ -a_{n-2} & 0 & & 0 \\ \vdots & & \ddots & 1 \\ -a_0 & 0 & & 0 \end{bmatrix} \right) \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} k_{n-1} - a_{n-1} \\ k_{n-2} - a_{n-2} \\ \vdots \\ k_0 - a_0 \end{bmatrix} y.$$

Replacing the last equality in (5.6) we finally get:

$$\begin{aligned}
x &= (Is - A_0)^{-1} \begin{bmatrix} k_{n-1} - a_{n-1} \\ k_{n-2} - a_{n-2} \\ \vdots \\ k_0 - a_0 \end{bmatrix} [y] + (Is - A_0)^{-1} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ b_m \\ \vdots \\ b_0 \end{bmatrix} [u] = \\
&= \sum_{i=0}^{n-1} (k_i - a_i) (Is - A_0)^{-1} e_{n-i} [y] + \sum_{j=0}^m b_j (Is - A_0)^{-1} e_{m-j} [u],
\end{aligned}$$

where $e_i^T = [0, \dots, 0, 1, 0, \dots, 0]$. Rewrite the latter as

$$x = \sum_{i=0}^{n-1} (k_i - a_i) (Is - A_0)^{-1} e_{n-i} [y] + \sum_{j=0}^m b_j (Is - A_0)^{-1} e_{m-j} [u],$$

or using notation of θ :

$$x = \sum_{i=0}^{n-1} \theta_{i+1} (Is - A_0)^{-1} e_{n-i} [y] + \sum_{j=0}^m \theta_{j+1+n} (Is - A_0)^{-1} e_{m-j} [u], \quad (5.7)$$

It is seen from (5.7) that it is enough to measure the input and the output variable to parameterize x for the plant (5.5) presented in canonical form.

It worth noting that expression (5.7) can be reduced to the form of Luenberger observer very well known in classic control theory [1, 2].

For the sake of simplicity of simulation in Matlab/Simulink environment the transfer vectors functions $(Is - A_0)^{-1} e_{n-i}$ can be represented as the outputs of the block “State-Space” from library “Linear”, where the parameters A, b, C, d are defined by A_0, e_{n-i} , identity matrix of dimension $n \times n$ (MatLab function “eye”) and zero vector of dimension $n \times 1$, respectively.

The order of the work

1. Using the conditions given in Table 5 build the plant (5.3) in the form (5.4). Provide the possibility to change the initial conditions $y^{(i)}(0), i = \overline{0, n-1}$ in (5.3). Calculate θ in (5.4) and build it.

Plot the outputs y of (5.3) and (5.4) for zero and nonzero initial conditions (5.3).

2. Using the conditions given in Table 5 build the plant (5.3) in the form (5.5). Provide the possibility to change the initial conditions $x_i(0), i = \overline{1, n}$. Build the parameterized model (5.7).

Plot the states x_1 , x_2 of (5.5) and (5.4) for zero and nonzero initial conditions.

3. Make conclusions.

Table 5 Parameters of the plant and the filters and the input signals

#	Plant parameters ($n = 2$)				Filter parameters		Input
	a_1	a_0	b_1	b_0	k_1	k_0	
1	1	1	1	2	0,2	0,01	$u = \sin t + 0,5 \cos 2t$
2	2	1	1	3	$\sqrt{0,02}$	0,01	
3	3	1	1	4	0,4	0,04	
4	4	1	1	5	$\sqrt{0,08}$	0,04	
5	5	1	1	6	0,6	0,09	
6	1	2	1	7	$\sqrt{0,18}$	0,09	
7	2	2	1	8	0,8	0,16	
8	3	2	1	9	$\sqrt{0,32}$	0,16	
9	4	2	2	3	1	0,25	
10	5	2	2	4	$\sqrt{0,5}$	0,25	
11	1	3	2	5	1,2	0,36	
12	2	3	2	6	$\sqrt{0,72}$	0,36	
13	3	3	2	7	1,6	0,64	
14	4	3	2	8	$\sqrt{1,28}$	0,64	
15	5	3	2	9	2	1	
16	1	4	3	4	$\sqrt{2}$	1	
17	2	4	3	5	4	4	
18	3	4	3	6	$2\sqrt{2}$	4	
19	4	4	3	7	6	9	
20	5	4	3	8	$3\sqrt{2}$	9	
21	1	5	3	9	8	16	
22	2	5	4	5	$4\sqrt{2}$	16	
23	3	5	4	6	1	0,25	
24	4	5	4	7	$\sqrt{0,5}$	0,25	
25	5	5	4	8	1,2	0,36	
26	1	6	4	9	$\sqrt{0,72}$	0,36	
27	2	6	5	6	1,6	0,64	
28	3	6	5	7	$\sqrt{1,28}$	0,64	
29	4	6	5	8	2	1	
30	5	6	5	9	$\sqrt{2}$	1	

Problems and questions

1. Parameterize the output of the discrete plant

$$y(k+n) + a_{n-1}(k+n-1) + \dots + a_0 y(k) = b_m u(k+m) + b_{m-1} u(k+m-1) + \dots + b_0 u(k),$$

where k is the sampling time.

2. Using (5.4) parameterize the first time derivative of y .

3. Show that if the initial conditions in (5.3) are not zero, then the parameterization is of the following form:

$$y = \theta^T \omega + \sigma_{y1}, \quad (5.8)$$

where σ_{y1} is the exponentially decaying term.

4. Show that if the initial conditions in (5.5) отличны от нуля (т.е. Допущение, принятое в работе, ослабляется), are not zero, then the parameterization has the following form:

$$x = \sum_{i=0}^{n-1} (k_i - a_i) (sI - A_0)^{-1} e_{n-i} [y] + \sum_{j=0}^m b_j (sI - A_0)^{-1} e_{m-j} [u] + \sigma_x,$$

where σ_x is the exponentially decaying vector satisfying $\dot{\sigma}_x = A_0 \sigma_x$.

5. Parameterize the first time derivative of x for the second order system given in the work with one of the set of parameters from Table 5.

6. Represent the output variable of the plant

$$\ddot{y} + \theta_1 \dot{y} + \theta_2 y = \theta_3 \ddot{u} + \theta_4 \dot{u} + \theta_5 u,$$

in the form $y = \omega_0 + \theta^T \omega$, where $\theta = [\theta_1, \dots, \theta_5]^T$ is the vector of plant parameters, $\omega = [\omega_1, \dots, \omega_5]^T$ is the vector of measurable functions, ω_0 is a measurable function. It is assumed that in the plant the variables y and u are accessible for measurement only.

7. Represent the output variable of the pendulum model

$$\ddot{y} + \theta_1 \dot{y} + \theta_2 \sin(y) = \theta_3 u,$$

in the form $y = \omega_0 + \theta^T \omega$, where $\theta = [\theta_1, \theta_2, \theta_3]^T$ is the vector of plant parameters, $\omega = [\omega_1, \omega_2, \omega_3]^T$ is the vector of measurable functions, ω_0 is a measurable function. It is assumed that in the plant the variables y and u are accessible for measurement only.

8. Represent the state variable of the plant

$$\begin{aligned} \dot{x}_1 &= \theta_1 x_1 + x_2, \\ \dot{x}_2 &= \theta_2 x_1 + x_1^2 + \theta_3 u, \\ y &= x_1, \end{aligned}$$

in the form $x = \omega_0 + \theta_1\omega_1 + \theta_2\omega_2 + \theta_3\omega_3$, where $\theta_1, \theta_2, \theta_3$ are the plant parameters, $\omega_0, \omega_1, \omega_2, \omega_3$ are the measurable functions. It is assumed that in the plant the variables y and u are measurable only.

9. How would the properties of the parameterized model (5.8) of the second order change if in the we increased the parameter ω of the polynomial

$$K(s) = s^2 + 2\omega s + \omega^2,$$

in 10 times?

10. Represent the output variable of the oscillating block

$$y = \frac{k}{T^2 s^2 + 2\zeta T s + 1} [u],$$

in the form $y = \omega_0 + \theta^T \omega$, where k is the gain, T is the time constant, ζ is the damping coefficient such that $0 < \zeta < 1$, $\theta \in R^{3 \times 1}$ is the vector depending on k , T , ζ , $\omega \in R^{3 \times 1}$ is the vector of some measurable functions, ω_0 is a measurable function. It is assumed that in the block the variables y and u are measurable only.

Работа #6. ADAPTIVE OBSERVER DESIGN FOR LTI PLANTS

Problem statement. Consider asymptotically stable plant

$$\begin{cases} \dot{x} = Ax + bu, & x(0), \\ y = c^T x, \end{cases} \quad (6.1)$$

with unmeasurable state vector x and the measurable input u and the output y . Let the plant matrices be presented in canonical (observable) form

$$A = \begin{bmatrix} -a_{n-1} & 1 & \cdots & 0 \\ -a_{n-2} & 0 & & 0 \\ \vdots & & \ddots & 1 \\ -a_0 & 0 & & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ b_m \\ \vdots \\ b_0 \end{bmatrix}, \quad c^T = [1 \ 0 \ \cdots \ 0],$$

where $a_i, i = \overline{0, n-1}, b_j, j = \overline{0, m}$ are unknown coefficients.

The problem is to design an observer providing the boundedness of all the signal in the closed-loop system and generating $\hat{x}$ such that under the persistent excitation condition (3.14) the following equality holds

$$\lim_{t \rightarrow \infty} \|x(t) - \hat{x}(t)\| = 0. \quad (6.2)$$

It is supposed that the adaptive observer designed simultaneously estimates the unknown parameters of the plant and generates $\hat{x}$.

In general case the class of plants, plant (6.1) belongs to, is restricted by the following assumption called *matching condition*.

Assumption (matching condition). For some n -dimensional vector $\bar{\theta}$ the matrices A, c^T and A_0 are related via the following equality:

$$A_0 = A + \bar{\theta}C.$$

It can be shown (see work #5) that for the plant considered

$$\bar{\theta} = \begin{bmatrix} k_{n-1} - a_{n-1} \\ k_{n-2} - a_{n-2} \\ \vdots \\ k_0 - a_0 \end{bmatrix}.$$

Problem solution. The problem is resolved in two steps. First step is the identification of the plant parameters, while the second step is to design the observer in with the unknown parameters will be replaced by their estimates.

Starting from the parameters identification we use parameterization of the output variable (5.4). Replacing the parameters θ by their estimates $\hat{\theta}$ in (5.4) we design the adjustable model of the plant:

$$\hat{y} = \hat{\theta}^T \omega, \quad (6.3)$$

where $\hat{y}$ is the estimate of y . Introduce the error of identification

$$\varepsilon = y - \hat{y}$$

and taking into account (5.4) get:

$$\varepsilon = \tilde{\theta}^T \omega, \quad (6.4)$$

where $\tilde{\theta} = \theta - \hat{\theta}$ is the vector of parametric errors. The last equality represents the static error model that can be used for design of the following adaptation algorithm based on standard Lyapunov function $V = \tilde{\theta}^T \tilde{\theta} / 2\gamma$:

$$\dot{\tilde{\theta}} = \gamma \omega \varepsilon, \quad (6.5)$$

where $\gamma > 0$ is the adaptation gain.

Indeed, calculation of $\dot{V}$ yields:

$$\dot{V} = \frac{1}{\gamma} \tilde{\theta}^T \dot{\tilde{\theta}} = -\frac{1}{\gamma} \tilde{\theta}^T \dot{\tilde{\theta}}.$$

Replacing (6.5) in the last equality we get nonpositive time derivative

$$\dot{V} = -\frac{1}{\gamma} \tilde{\theta}^T \gamma \omega \varepsilon = -\varepsilon^2 < 0. \quad (6.6)$$

In the second step we replace in (5.7) the unknown parameters θ by the estimates $\hat{\theta}$ generated by (6.5) and obtain adjustable observer

$$\hat{x} = \sum_{i=0}^{n-1} \hat{\theta}_{i+1} (sI - A_0)^{-1} e_{n-i} [y] + \sum_{j=0}^m \hat{\theta}_{j+1+n} (sI - A_0)^{-1} e_{m-j} [u]. \quad (6.7)$$

If the functions ω and $\dot{\omega}$ are bounded (the condition depends on the input u) it follows from (6.6) that the adaptive observer consisting of adjustable observer (6.7), adaptation algorithm (6.5), error $\varepsilon = y - \hat{y}$ and adjustable parameterized model (6.3) has the following properties:

- P.1. all the signals in the observer are bounded;
- P.2. identification error ε approaches zero asymptotically;

P.3. Observation error $x - \hat{x}$ and parametric error $\tilde{\theta}$ approach zero exponentially fast, if ω satisfies the persistent excitation condition (3.14) (see [6, 7]). Condition (3.14) depends on the number of harmonics (spectral lines) in the signal u .

The order of the work

1. Based on the result obtained in work #5 build and simulate adaptive observer (6.3), (6.5), (6.6). Adaptation gain γ is selected experimentally. Plot the norm $\|x - \hat{x}\|$ and the norm of the vector $\tilde{\theta}$ separately.

2. Repeat the experiment for $u = 10\sin t + 5\cos 2t + 4\cos 4t + 3\cos 8t$.

3. Make a conclusion.

Problems and questions

1. It is known from adaptive control theory [4, 6, 7] that, in accordance to so called MIT rule, adaptation algorithms can be derived via steepest decent approach:

$$\dot{\tilde{\theta}} = \gamma \underset{\tilde{\theta}}{\text{grad}} \{V(\varepsilon, \tilde{\theta})\}. \quad (6.7)$$

where V can be interpreted as a performance index to be minimized. Show that if $V = \varepsilon^2 / 2$ (where ε is defined by (6.4)), then adaptation algorithm (6.7) can be presented in the form (6.5).

2. Based on (6.7) and the selected index

$$V = |\varepsilon|,$$

where ε is defined by (6.4), design an adaptation algorithm and make experiments in accordance with item#1 and 2 of this work.

3. Design an adaptive observer of the time derivative of y .

4. How would the design of adaptive observer be changed if the plant (6.1) presented in a noncanonical form?

5. Design the adaptive observer for the plant

$$\dot{x}_1 = x_2 - x_1,$$

$$\dot{x}_2 = \theta u - x_1,$$

$$y = x_1,$$

where $\theta = 3$ is unknown parameter. It is assumed that y and u are measurable only. In the design procedure minimize the dynamic orders of the filters and the

adaptation algorithm. Make simulation experiments with $u = 5\sin t + \cos 0,5t$ and experimentally selected filter coefficients and the adaptation gain.

6. Will the adaptive state observer provide the objective (6.2) if the plant is stable, observable and is given by equation

$$\begin{aligned}\dot{x}_1 &= -a_1 x_1 + x_2, \\ \dot{x}_2 &= -a_0 x_1 + b_0 u, \\ y &= x_1,\end{aligned}$$

where a_0, a_1, b_0 are unknown parameters, $u = 1$? Please clarify.

7. How are the properties of adaptive observer changed if we apply robust σ -modification of the adaptation algorithm

$$\dot{\hat{\theta}} = -\sigma \hat{\theta} + \gamma \omega \varepsilon,$$

where σ is a positive constant? Please clarify.

8. Would be the objective (6.2) provided by the adaptive observer if:

$$\begin{aligned}\text{a) } A &= \begin{bmatrix} -5 & 1 \\ -6 & 0 \end{bmatrix}, b = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, C = [1 \ 0]; \\ \text{б) } A &= \begin{bmatrix} -5 & 1 \\ -6 & 0 \end{bmatrix}, b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C = [1 \ 0]; \\ \text{в) } A &= \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}, b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C = [3 \ 1]?\end{aligned}$$

Please clarify.

9. Design an adaptive observer for the plant

$$\begin{aligned}\dot{x}_1 &= -x_1 + x_2, \\ \dot{x}_2 &= \theta x_1 + u, \\ y &= x_1,\end{aligned}$$

where $\theta = -4$ is unknown parameter. It is assumed that y and u are measurable only. In the design procedure minimize the dynamic orders of the filters and the adaptation algorithm. Make simulation experiments with $u = 2\text{sign}(\sin t) + 3$ and experimentally selected filter coefficients and the adaptation gain.

10. Design the adaptive observer of the pendulum angle speed $\dot{y}$ if the model of the pendulum has the following form:

$$\ddot{y} + \theta_1 \dot{y} + \theta_2 \sin(y) = \theta_3 u,$$

where $\theta_1, \theta_2, \theta_3$ are unknown parameters depending on the pendulum geometry, mass, inertia and friction. It is assumed that the variables y and u are measurable only.

References

1. C.-T. Chen, Linear Systems. Theory and Design, 3rd edn. Oxford University Press, New York, Oxford, 1999
2. R.C. Dorf, Modern Control Systems, 10th ed. Reading, MA: Addison-Wesley, 2004.
3. Ioannou P.A., Kokotovic P. V. Instability analysis and improvement of robustness of adaptive control // Automatica. — 1984. Vol. 20. — P. 583-594.
4. Ioannou P.A. and Sun J. Robust adaptive control — California: Prentice-Hall, 1996. — 848 p.
5. Khalil H. K., Nonlinear Systems, 3rd Edition, Prentice Hall, Upper Saddle River, 2002. — 766 p. (Русский пер.: Халил Х.К. Нелинейные системы. — М. — Ижевск: НИЦ "Регулярная и хаотическая динамика", Институт компьютерных исследований, 2009. — 832 с.)
6. Narendra K.S. and Annaswamy A.M. Stable Adaptive Systems. — Englewood Cliffs. N.J.: Prentice-Hall, 1989. — 495 p.
7. Sastry S. and Bodson M., Adaptive Control: Stability, Convergence and Robustness. — Englewood Cliffs. N.J.: Prentice-Hall, 1989. — 377 p.